u6-semiconductors

n01-intrinsic_carrier_concentration

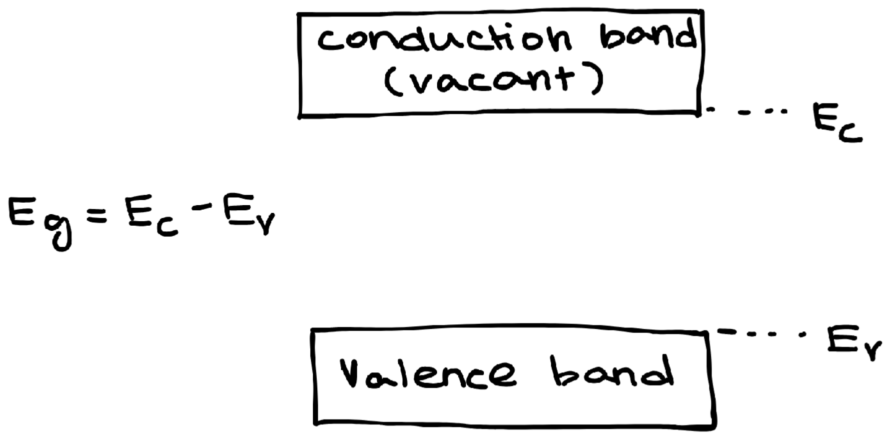
In an intrinsic semiconductor the breaking of each covalent bond creates equal no of electrons in conduction band and holes in valency band.

- If n is the electron carrier concentration in conduction band i.e. no of electrons per unit volume and p is hole carrier concentration and n_i is the intrinsic carrier concentration,

$$n = N_c e^{-\frac{(E_c - E_F)}{KT}} \dots (i)$$

$$p = N_v e^{-\frac{(E_F - E_v)}{KT}} \dots (ii)$$

- where, N_c, N_v are pseudo constants
- E_F is fermi level energy
- N_c and N_v are density of states in conduction and valency band respectively
- $K \to$ Boltzmann constant
- $T \to$ temp



For an intrinsic semiconductor

$$n_i^2 = n \times p$$

From (i) & (ii)

$$n_i^2 = N_c e^{-\frac{(E_c - E_F)}{KT}} \times N_v e^{-\frac{(E_F - E_v)}{KT}}$$

$$n_i^2 = N_c N_v e^{-\frac{(E_c - E_F + E_F - E_v)}{KT}}$$

$$n_i^2 = N_c N_v e^{-\frac{(E_c - E_v)}{KT}}$$

$$n_i^2 = N_c N_v e^{-\frac{E_g}{KT}}$$

$$n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2KT}} \ldots (iii)$$

- The above eqn shows that intrinsic carrier concentration is independent of fermi energy (E_F)
- Intrinsic carrier concentration is a function of E_g
- n_i depends on temperature

Variation of fermi level with temperature

n02-fermi_level

Fermi level

It indicates the probability of occupancy of energy levels in conduction and valency bands respectively.
In an intrinsic semiconductor since $n = p$, probability of occupancy of energy levels in conduction and valency band are equal. Therefore fermi energy level lies in the middle of the energy band gap.

$$n = p$$

$$N_c e^{-\frac{(E_c - E_F)}{KT}} = N_v e^{-\frac{(E_F - E_v)}{KT}}$$

$$e^{-\frac{(E_c - E_F)}{KT}} = e^{-\frac{(E_F - E_v)}{KT}}$$

taking ln on both sides

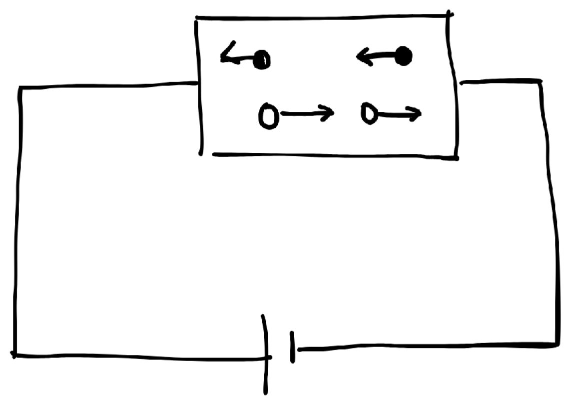
$$-\frac{(E_c - E_F)}{KT} = -\frac{(E_F - E_v)}{KT}$$

$$E_c - E_F = E_F - E_v$$

$$2E_F = E_c + E_v$$

$$E_F = \frac{E_c + E_v}{2}$$

- Therefore fermi energy level lies in the middle of the energy level.
- Consider an intrinsic semiconductor through which potential difference 'V' is applied as shown in figure.



- where μ is the mobility of charge carriers
- For electrons,

$$v_d = \mu_n E$$

- For holes,

$$v_d = \mu_p E$$

- Now current density for electrons density is given by

$$J_n = nev_d = ne\mu_n E \ldots (i)$$

$$J_p = pe\mu_p E \ldots (ii)$$

- The total current density is given by

$$J = J_n + J_p = ne\mu_n E + pe\mu_p E$$

$$J = e(n\mu_n + p\mu_p)E$$

- From ohms Law $J = \sigma E$

$$\sigma = \frac{J}{E} = e(n\mu_n + p\mu_p)$$

- For intrinsic semiconductor

$$n = p = n_i$$

$$\sigma = e(n_i\mu_n + n_i\mu_p)$$

$$\sigma = n_i e(\mu_n + \mu_p)$$

$$\sigma = e(\mu_n + \mu_p)\sqrt{N_c N_v} e^{-\frac{E_g}{2KT}}$$

- To find energy band gap
  where $A = e(\mu_n + \mu_p)(N_c N_v)^{1/2}$

$$\sigma = Ae^{-\frac{E_g}{2KT}}$$

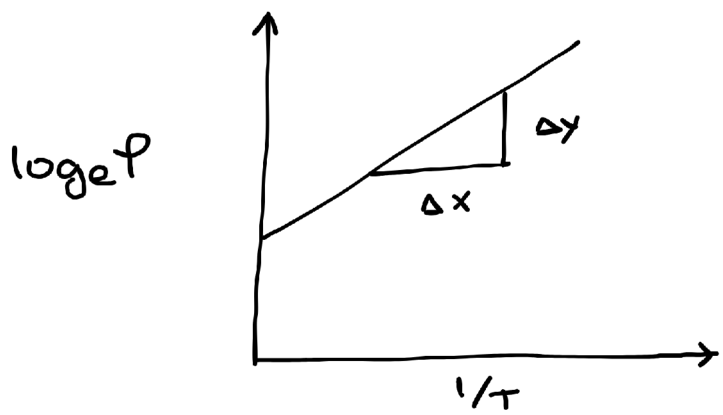
$$P = \frac{1}{\sigma}$$

$$P = \frac{1}{A}e^{\frac{E_g}{2KT}}$$

$$\rho = Be^{\frac{E_g}{2KT}}$$

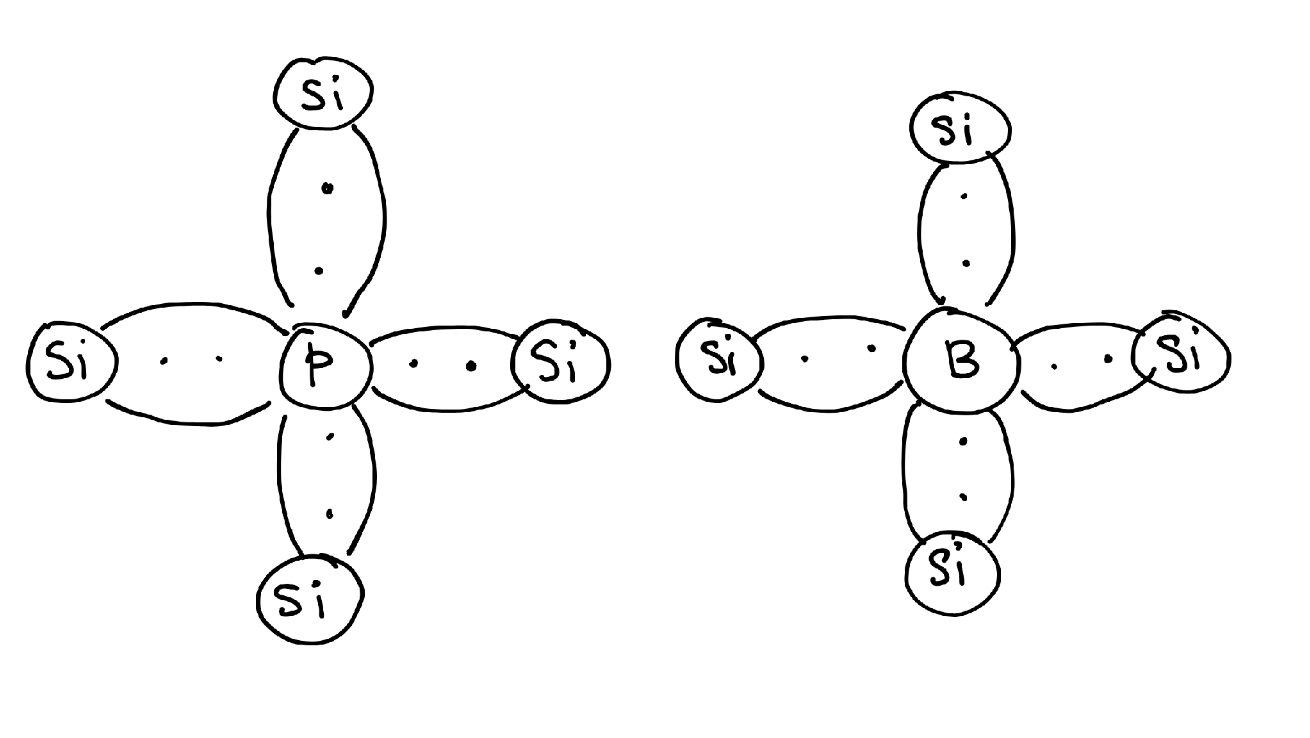
$$\ln\rho = \ln B + \frac{E_g}{2KT}$$

- This is an equation of straight line.
- A graph of $\ln\rho$ vs $\frac{1}{T}$ produces a straight line.



$$E_g = 2K(\text{slope})$$

n03-extrinsic_semiconductors



Fermi level in an extrinsic semiconductor
For n-type

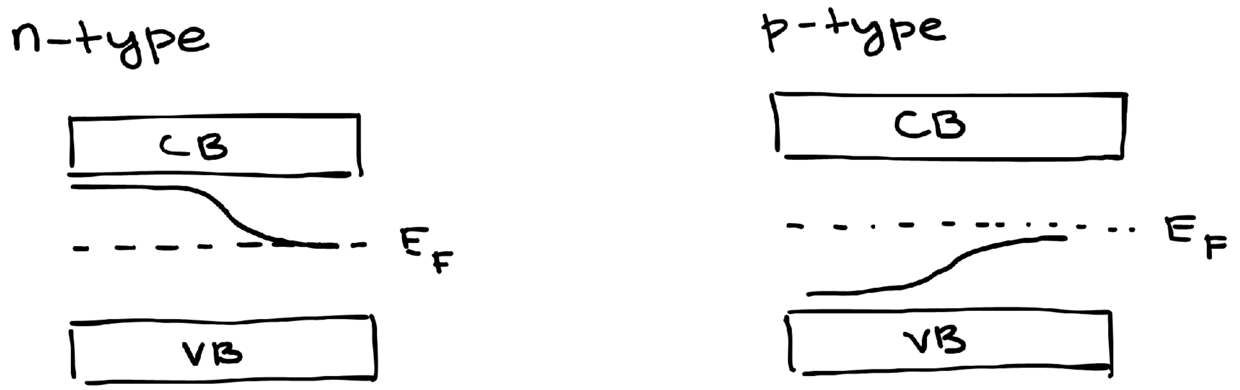
$$E_F = E_c - KT\ln(\frac{N_c}{N_D})$$

For p-type

$$E_F = E_v + KT\ln(\frac{N_v}{N_A})$$

where N_D is no of donor atoms
N_A is no of acceptor atoms

n04-variation_of_fermi_level_with_temperature



i) For an intrinsic semiconductor fermilevel is independent of temperature.
ii) For an n-type semiconductor fermi level shifts down towards middle position when 'T' increases. Then it works like an intrinsic semiconductor.
iii) For p-type semiconductor the fermi level shifts upward as temperature increases.

n05-law_of_mass_action

- It states that the product of majority carrier at a particular temperature is constant, which is equal to the square of intrinsic carrier concentration at that temperature.

For intrinsic semiconductor $n_i^2 = np$

n-type: $n_i^2 = n_n p_n$
where $n_n \rightarrow$ majority carriers (e's)
$p_n \rightarrow$ minority carriers (holes)

p-type $n_i^2 = n_p p_p$
$n_p =$ minority carriers (electrons)
$p_p =$ majority carriers (holes)

n06-drift

- Under influence of electric field the charge carriers are forced to move in different directions. This process is called drift.
- The velocity acquired by charge carriers is known as drift velocity.

$$v_d = \mu E$$

$$J = nev_d$$

$$J = ne\mu E$$

$$J = \sigma E$$

$$\sigma = ne\mu$$

n07-diffusion

- due to non uniform carrier concentration, the charge carriers move from high concentration region to lower concentration region.

n08-effective_mass

- the mass of electron when it is bound in a solid material is always different from that of the free electron mass.
- for some solids it was found to be high and for some solids it is low.
- Therefore mass of the electron determined experimentally when it is bound in a solid material is called effective mass of the electron and denoted as m^*.
- According to the De-Broglie hypothesis, each moving particle is associated with a group of waves having group velocity.

$$v_g = \frac{d\omega}{dk} \dots (i)$$

$$E = \hbar\omega = \hbar(2\pi\nu) = h\nu$$

$$\omega = \frac{E}{\hbar}$$

$$v_g = \frac{d}{dk}(\frac{E}{\hbar})$$

$$v_g = \frac{1}{\hbar}\frac{dE}{dk}$$

differentiating with respect to t

$$\frac{dv_g}{dt} = \frac{1}{\hbar}\frac{d}{dt}(\frac{dE}{dk})$$

$$\frac{dv_g}{dt} = \frac{1}{\hbar}[\frac{d^2E}{dk^2}\frac{dk}{dt}]$$

$$a = \frac{dv_g}{dt} = \frac{1}{\hbar}(\frac{d^2E}{dk^2})(\frac{dk}{dt}) \dots (iii)$$

From De-Broglie hypothesis

$$P = \hbar k$$

$$F = \frac{dP}{dt} = \hbar\frac{dk}{dt} \dots (iv)$$

$$\frac{dk}{dt} = \frac{F}{\hbar}$$

Dividing (iii) by (iv)

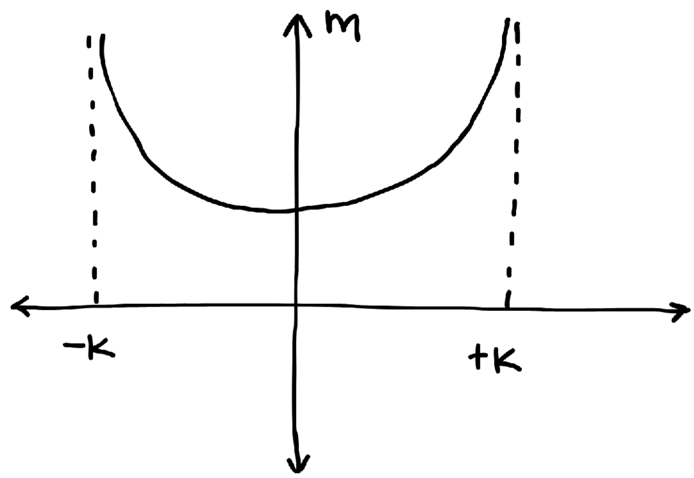
$$a = \frac{1}{\hbar}(\frac{d^2E}{dk^2})(\frac{F}{\hbar})$$

$$a = \frac{F}{\hbar^2}(\frac{d^2E}{dk^2})$$

$$F = [\frac{\hbar^2}{(d^2E/dk^2)}]a$$

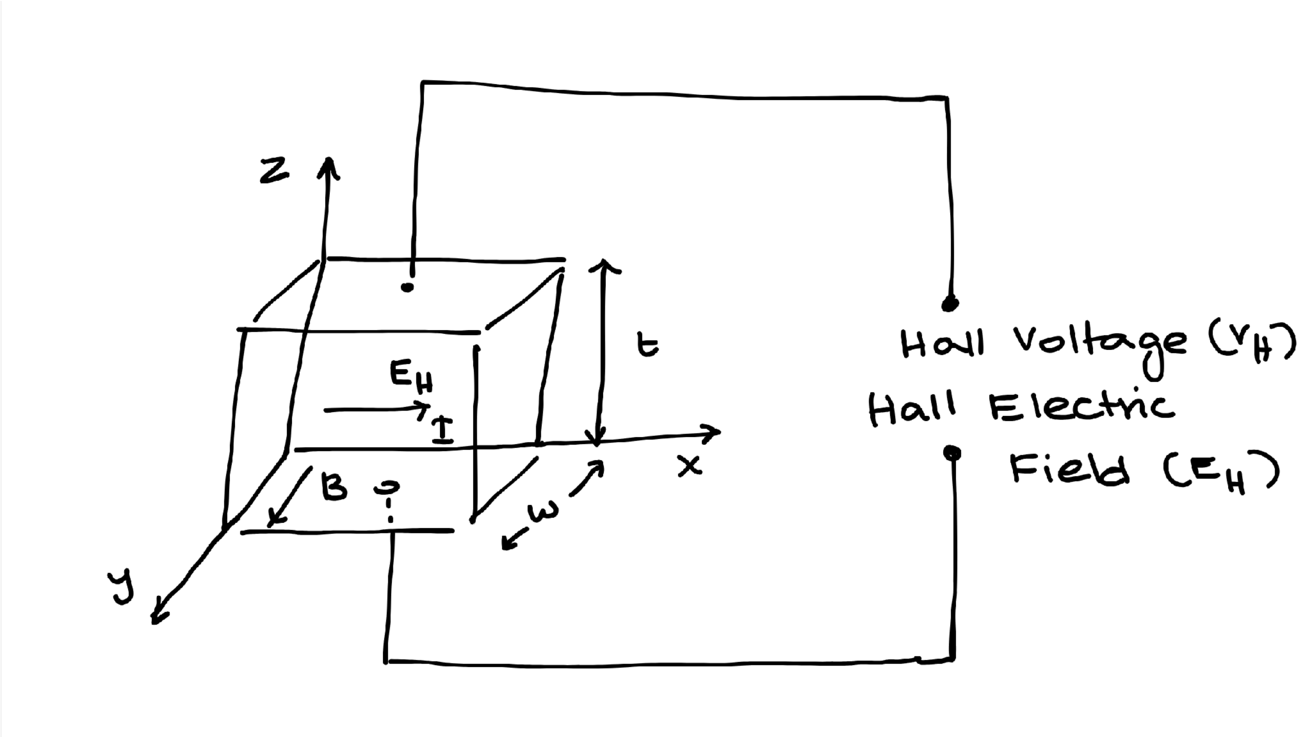
$$F = m^*a$$

$$m^* = \frac{\hbar^2}{(d^2E/dk^2)}$$



- According to the graph, we can say that there always exists a minimum mass for a particle.

n09-hall_effect



- When a current carrying metal or semiconductor is placed in a transverse magnetic field voltage developed normal to both current and magnetic field.
- This effect is known as Hall effect, the produced voltage is Hall voltage and electric field is called Hall electric field.
- Consider a semiconductor of thickness ´t´ and width ´w´ placed in a transverse magnetic field.
- Let I be the current passing in x-direction and B be y direction.
- Magnetic force is present which makes electrons occupy the bottom surface of the semiconductor.
- Thus the voltage is developed in z-direction and electric field is established.
- In steady state both forces are equal.

$$F_{mag} = F_{electric}$$

$$Bev = eE_H$$

$$E_H = Bv \ldots (i)$$

If V_H is the Hall voltage, then

$$E_H = \frac{V_H}{}$$

$$V_H = Bv \ldots (ii)$$

If J is the current density then

$$J = nev$$

$$v = \frac{J}{ne} = \frac{I}{Ane}$$

where A is the area of the slab $A = t$

$$v = \frac{I}{tne} \ldots (iii)$$

Substitute (iii) in (ii)

$$V_H = B(\frac{I}{tne})$$

$$V_H = \frac{BI}{net}$$

The Hall Coefficient R_H is defined as the $R_H = \frac{1}{ne}$

$$V_H = \frac{BIR_H}{t}$$

$$\sigma = ne\mu = \frac{\mu}{R_H}$$

$$\mu = \sigma R_H$$

Application

i) R_H is +ve p-type
R_H is -ve n-type
ii) Find carrier concentration
iii) Find mobility and conductivity
iv) Find no. of majority carriers
v) velocity of the majority carriers.